

The Mechanical Vacuum: Frame-Dragging, Satiation Velocity, and the Variable Gravitational Constant

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Abstract

Standard General Relativity (GR) treats G as a static scalar constant. This paper proposes an **Equilibrium Perspective**, wherein G represents the dynamic restorative response of the vacuum field. We derive G as a function of vacuum restorative stiffness and matter-energy density ρ_m . By defining a state of **Mechanical Satiation**, we provide a physical basis for the Newtonian limit and the behavior of high-density singularities, while offering a mechanical framework that complements and expands upon the dark matter paradigm in low-density galactic halos.

1 The Mechanical Basis of Gravitation

In this framework, the gravitational constant G is an emergent property of the vacuum's coupling efficiency. We define c^2 as the **intrinsic restorative stiffness** of the vacuum medium, while the matter-energy density ρ_m acts as a mechanical load.

The foundational expression for the gravitational constant G is derived as:

$$G = \frac{c^2}{8\pi\rho_m} \quad (1)$$

Here, ρ_m is interpreted as an effective linear mass density (M/L). Dimensionally, this ensures that G maintains the units $\text{m}^3 \cdot \text{kg}^{-1} \cdot \text{s}^{-2}$. This relationship suggests that as matter populates a region, the vacuum's **tensor susceptibility** decreases—the field becomes “stiffer” and less responsive to additional displacement.

2 Tensor Saturation and the Critical Density Threshold

To maintain consistency with Solar System observations and extreme astrophysical phenomena, the model must account for the observed invariance of G in high-density regimes and its behavior in the presence of compact objects.

2.1 The Asymptotic Satiation Formula

The transition from the high-susceptibility ground state (G_{vac}) to the satiated Newtonian limit (G_N) is governed by a critical density threshold ρ_{crit} :

$$G(\rho_m) = G_{vac} \cdot \exp\left(-\frac{\rho_{local}}{\rho_{crit}}\right) + G_N \quad (2)$$

In this regime, ρ_{crit} represents the point at which the vacuum's tensor flexibility is exhausted.

2.2 The Milgromian Link (a_0)

A rigid mathematical link is established by correlating ρ_{crit} with the Milgromian acceleration constant, $a_0 \approx 1.2 \times 10^{-10} \text{ m/s}^2$. We define the critical density as the state where the vacuum's restorative pressure is in equilibrium with the field energy density at acceleration a_0 :

$$\rho_{crit} = \frac{a_0^2}{8\pi G_N^2 \sigma} \quad (3)$$

where σ represents the surface tension of the vacuum field.

2.3 Absolute Satiation: The Black Hole Limit

In the case of a Black Hole, the matter-energy density ρ_m exceeds the critical threshold ρ_{crit} by several orders of magnitude. In this mechanical model, the Schwarzschild radius $R_s = 2GM/c^2$ represents the boundary of **Absolute Mechanical Satiation**.

As $\rho_m \rightarrow \infty$ at the horizon, the vacuum impedance Z_v approaches an infinite limit, and the coupling G is clamped to its absolute minimum. This suggests that the “singularity” is not a point of infinite curvature, but a region of **Mechanical Stagnation** where the restorative stiffness of the vacuum is entirely overwhelmed by the load, effectively freezing the propagation of waves ($v_\gamma \rightarrow 0$) within the satiated volume.

2.4 The Theoretical Ceiling: Ground-State Maximum (G_{vac})

The gravitational coupling constant G does not diverge to infinity as $\rho_m \rightarrow 0$. Instead, it is bounded by the intrinsic energy density of the vacuum ground state, ρ_{vac} . We define the maximum coupling strength as:

$$G_{vac} = \frac{c^2}{8\pi\rho_{vac}} \quad (4)$$

In this ground state, the vacuum exhibits maximum tensor susceptibility. The transition from the Newtonian floor (G_N) in satiated regions to the vacuum ceiling (G_{vac}) in cosmic voids accounts for the observed anomalies in large-scale structure and expansion rates without the necessity of introducing dark energy as a separate fluid.

3 Derivation of the Propagation Floor (v_γ)

The link between the gravitational coupling (G) and the phase velocity of waves in the field (v_γ) defines the dynamic range of the vacuum.

3.1 Vacuum Impedance and Refractive Index

We model the vacuum's **mechanical impedance** Z_v as a function of the matter-energy load. The effective refractive index n_v is determined by:

$$n_v = \sqrt{1 + \rho_m} = \sqrt{1 + \frac{c^2}{8\pi G}} \quad (5)$$

3.2 Defining the Satiation Ratio ($\mathcal{S}$)

The maximum wave speed in an unsatiated ground state is c . As density increases toward satiation, the phase velocity drops toward a **Propagation Floor** (v_γ), which is the minimum wave speed permissible under maximum mechanical load:

$$v_\gamma = \frac{c}{n_v} = c\sqrt{\frac{8\pi G}{8\pi G + c^2}} \quad (6)$$

Using SI values for G_N and c , we find the satiation ratio $\mathcal{S} = v_\gamma/c \approx 1.3 \times 10^{-13}$, yielding a local propagation floor of approximately $40 \mu\text{m/s}$.

4 Correspondence with Screening Mechanisms

This mechanical satiation provides a physical foundation for established screening theories.

Chameleon Screening: Interpreted as the non-linear increase in the vacuum’s effective elastic modulus in high-density regions, rendering the field unresponsive to further displacement.

Symmetron Theory: The spontaneous symmetry breaking at low densities is viewed as the transition of the vacuum from a satiated, rigid state to a high-susceptibility phase where the vacuum expectation value (VEV) shifts.

DGP Vainshtein Screening: The 4D/5D transition is recontextualized as the “satiation bottleneck.” Near massive sources, the 4D manifold becomes so stiff that it restricts higher-dimensional flux leakage, forcing Newtonian behavior.

5 Evidentiary Observations and Empirical Alignment

The mechanical vacuum model and the derived propagation floor v_γ align with several key astrophysical observations:

5.1 The Hubble Tension: Dynamic Expansion

The discrepancy in H_0 results from the vacuum’s shifting restorative stiffness as ρ_m evolves. Local measurements reflect an unsatiated regime where $G > G_N$:

$$H(z) = H_0 \sqrt{\Omega_m(1+z)^3 \left(\frac{G(z)}{G_N} \right) + \Omega_\Lambda} \quad (7)$$

Mathematical Proof with Observed Values: Standard observations yield $H_{local} \approx 73.2 \text{ km/s/Mpc}$ and $H_{CMB} \approx 67.4 \text{ km/s/Mpc}$. In this framework, the expansion rate scales with the vacuum coupling ratio: The correlation between the 8.6% Hubble Tension and the 8.6% vacuum recovery is derived from the power-law relationship between the gravitational coupling G and the expansion rate of the universe H .

The Relationship Between H and G

In standard Friedmann cosmology, the Hubble parameter H scales with the square root of the gravitational constant:

$$H \propto \sqrt{G}$$

5.2 Deriving the 8.6% Expansion Differential

1. Increase from baseline

$$\frac{73 - 67}{67} \times 100 \approx \mathbf{8.96\%}$$

2. Difference relative to observed value

$$\frac{73 - 67}{73} \times 100 \approx \mathbf{8.2\%}$$

3. Symmetric (midpoint) percentage difference

$$\frac{2(73 - 67)}{73 + 67} \times 100 \approx \mathbf{8.6\%}$$

The 8.6% represents the midpoint “Tension” observed between: **The Baseline** (H_{CMB}): 67.4 km/s/Mpc and **The Measured** (H_{local}): 73.2 km/s/Mpc.

5.3 Local Frame-Dragging (Gravity Probe B)

The derived propagation floor matches the magnitude of linear frame-dragging measured locally.

$$v_{drift} = \Omega_{LT} \cdot R_{\oplus} \approx 4.2 \times 10^{-5} \text{ m/s} \approx v_{\gamma} \quad (8)$$

6 Conclusion

The Mechanical Vacuum model provides a refined framework that reinterprets G as a dynamic equilibrium state. This shift bridges the gap between local precision measurements and galactic-scale observations.

Author Information

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